



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

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**2024 Gauss Grade 8
Solutions**

1. How many 5 cent coins are needed to make 25 cents?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Source: 2024 Gauss Grade 8 #1

Primary Topics: Number Sense

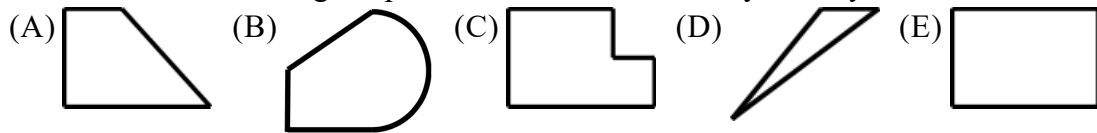
Secondary Topics: Operations | Counting

Answer: E

Solution:

The number of 5 cent coins needed to make 25 cents is $\frac{25 \text{ cents}}{5 \text{ cents}} = 5$.

2. Which of the following shapes has a vertical line of symmetry?



Source: 2024 Gauss Grade 8 #2

Primary Topics:

Secondary Topics: Polygons | Transformations

Answer: E

Solution:

If a figure has a vertical line of symmetry, then when it is reflected in (or folded along) this vertical line, the two halves of the figure are identical to one another. Of the given shapes, such a vertical line is only possible for (E).

3. Which of the following numbers is largest?

- (A) 0.58 (B) 1.32 (C) 0.97 (D) 1.03 (E) 0.12

Source: 2024 Gauss Grade 8 #3

Primary Topics: Number Sense

Secondary Topics: Decimals | Digits

Answer: B

Solution:

Of the given numbers, only 1.32 and 1.03 are greater than 1. Since the tenths digit of 1.32, namely 3, is greater than the tenths digit of 1.03, which is 0, then 1.32 is the largest number in the list.

4. 50% of n is 2024. The value of n is

- (A) 2074 (B) 24 (C) 50 (D) 4048 (E) 4042

Source: 2024 Gauss Grade 8 #4

Primary Topics: Number Sense

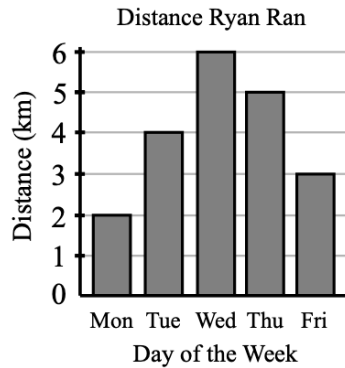
Secondary Topics: Percentages

Answer: D

Solution:

If 50% of n is 2024, then half of n is 2024, and so n is equal to 2 times 2024, which is equal to 4048. (We may confirm that 50% of 4048, or half of 4048, is 2024 as required.)

5. Ryan recorded the distance, in kilometres, that he ran on each day from Monday to Friday, as shown.



The total distance that Ryan ran over the five days is

- (A) 14 km (B) 16 km (C) 18 km (D) 20 km (E) 22 km

Source: 2024 Gauss Grade 8 #5

Primary Topics:

Secondary Topics: Counting | Graphs

Answer: D

Solution:

Reading from the graph, Ryan ran 2 km on Monday, 4 km on Tuesday, 6 km on Wednesday, 5 km on Thursday, and 3 km on Friday.

Thus, the total distance that Ryan ran over the five days is $2 + 4 + 6 + 5 + 3 = 20$ km.

6. When the number 11 is increased by 2 and the result is then multiplied by 3, the final result is

- (A) 35 (B) 39 (C) 28 (D) 25 (E) 363

Source: 2024 Gauss Grade 8 #6

Primary Topics: Algebra and Equations

Secondary Topics: Equations Solving | Operations

Answer: B

Solution:

When 11 is increased by 2, the result is 13.

When 13 is then multiplied by 3, the final result is $13 \times 3 = 39$.

7. The value of a that satisfies the equation $15 + a = 10$ is

- (A) -10 (B) -5 (C) 0 (D) 5 (E) 10

Source: 2024 Gauss Grade 8 #7

Primary Topics: Algebra and Equations

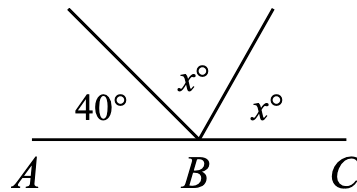
Secondary Topics: Equations Solving | Operations

Answer: B

Solution:

If $15 + a = 10$, then $a = 10 - 15$ and so $a = -5$.

8. In the diagram, $\angle ABC$ is a straight angle.



The value of x is

- (A) 80 (B) 65 (C) 75 (D) 70 (E) 60

Source: 2024 Gauss Grade 8 #8

Primary Topics: Geometry and Measurement

Secondary Topics: Angles | Equations Solving

Answer: D

Solution:

Since the measure of a straight angle is 180° , then $40^\circ + x^\circ + x^\circ = 180^\circ$.

Solving this equation, we get $40 + x + x = 180$ or $2x = 140$, and so $x = 70$.

9. In a drawer, the ratio of the number of spoons to the number of forks is 1 : 2. The total number of spoons and forks in the drawer *cannot* be equal to
- (A) 12 (B) 6 (C) 18 (D) 10 (E) 3

Source: 2024 Gauss Grade 8 #9

Primary Topics: Number Sense

Secondary Topics: Fractions/Ratios | Counting

Answer: D

Solution:

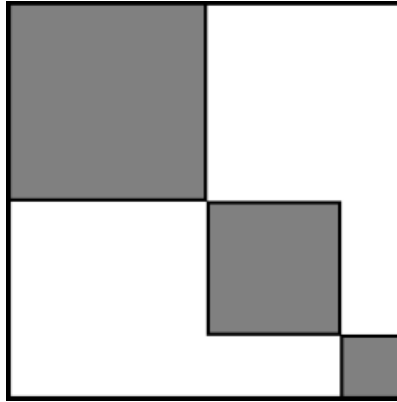
The ratio of the number of spoons to the number of forks is 1 : 2, which means that for each spoon in the drawer, there are 2 forks in the drawer.

That is, the spoons and forks in the drawer can be separated into groups of 3 (1 spoon and 2 forks), and thus the total number of spoons and forks in the drawer must be a multiple of 3.

Each of the answers 12, 6, 18, and 3 is a multiple of 3.

The only answer given that is not a multiple of 3 is 10, and so the total number of spoons and forks in the drawer cannot be 10.

10. In the diagram, a square with side length 6 is partially shaded.



The largest shaded region is a square with side length 3. The other two shaded regions are squares with side lengths 2 and 1. What is the total area of the unshaded region?

- (A) 12 (B) 18 (C) 22 (D) 24 (E) 30

Source: 2024 Gauss Grade 8 #10

Primary Topics: Geometry and Measurement

Secondary Topics: Area | Quadrilaterals

Answer: C

Solution:

The large square with side length 6 has area $6 \times 6 = 36$.

The shaded squares with side lengths 3, 2 and 1, have areas $3 \times 3 = 9$, and $2 \times 2 = 4$, and $1 \times 1 = 1$, respectively.

The total area of the shaded regions is $9 + 4 + 1 = 14$.

The total area of the unshaded region is equal to the area of the shaded regions subtracted from the area of the large square, or $36 - 14 = 22$.

11. In the sequence $1, 3, 4, 7, \dots$, each number beginning with the 4 is the sum of the two numbers before it. This means that the next number in the sequence is $4 + 7 = 11$. The smallest number greater than 100 that appears in the sequence is
(A) 133 (B) 199 (C) 113 (D) 101 (E) 123

Source: 2024 Gauss Grade 8 #11

Primary Topics: Algebra and Equations

Secondary Topics: Patterning/Sequences/Series | Counting

Answer: E

Solution:

Continuing the sequence, we get 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, and so the smallest number greater than 100 that appears in the sequence is 123.

12. The number 385 has three prime factors. The sum of these prime factors is
(A) 21 (B) 26 (C) 25 (D) 23 (E) 22

Source: 2024 Gauss Grade 8 #12

Primary Topics: Number Sense

Secondary Topics: Factoring | Prime Numbers

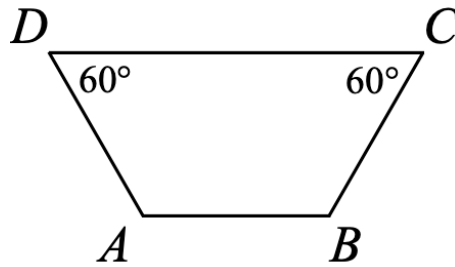
Answer: D

Solution:

Since 385 has a units digit of 5, then it is divisible by 5. Dividing, we get $\frac{385}{5} = 77$.

Since $77 = 7 \times 11$, then the three prime factors of 385 are 5, 7 and 11, and their sum is $5 + 7 + 11 = 23$.

13. Trapezoid $ABCD$ can be divided into three equilateral triangles.



If the perimeter of the trapezoid is equal to 840cm , what is the length of AB ?

- (A) 120 cm (B) 140 cm (C) 168 cm (D) 25 cm (E) 210 cm

Source: 2024 Gauss Grade 8 #13

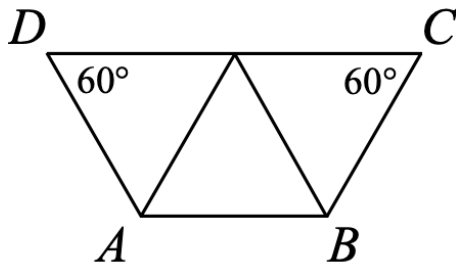
Primary Topics: Geometry and Measurement

Secondary Topics: Quadrilaterals | Perimeter | Triangles

Answer: C

Solution:

We begin by splitting $ABCD$ into three equilateral triangles, as shown. (Confirm for yourself that this is the only way to do this.)



Since the equilateral triangle in the middle shares a side with each of the two other equilateral triangles, then the sides of all three equilateral triangles are equal in length.

The perimeter of $ABCD$ is made up of 5 side lengths of these equilateral triangles.

Since the perimeter of $ABCD$ is 840 cm , then each side length of these equilateral triangles is

$$\frac{840\text{ cm}}{5} = 168\text{ cm}.$$

Since AB is a side of an equilateral triangle, then $AB = 168\text{ cm}$.

14. A container of ice cream can make 6 cones or it can make 4 sundaes. If 5 such containers of ice cream are used to make 12 cones, what is the greatest number of sundaes that can be made with the ice cream that remains?

(A) 4 (B) 8 (C) 12 (D) 16 (E) 20

Source: 2024 Gauss Grade 8 #14

Primary Topics: Algebra and Equations

Secondary Topics: Equations Solving | Inequalities

Answer: C

Solution:

One container of ice cream can make 6 cones.

Thus, 2 containers of ice cream can make $2 \times 6 = 12$ cones.

Since 12 cones were made from 5 containers of ice cream, and it takes 2 containers of ice cream to make 12 cones, then $5 - 2 = 3$ containers of ice cream remain after the 12 cones are made.

One container of ice cream can make 4 sundaes.

Thus, the remaining 3 containers of ice cream can make $3 \times 4 = 12$ sundaes.

15. When a positive integer n is divided by 10, the remainder is 8. When n is divided by 5, the remainder is

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Source: 2024 Gauss Grade 8 #15

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Divisibility | Operations | Expressions

Answer: D

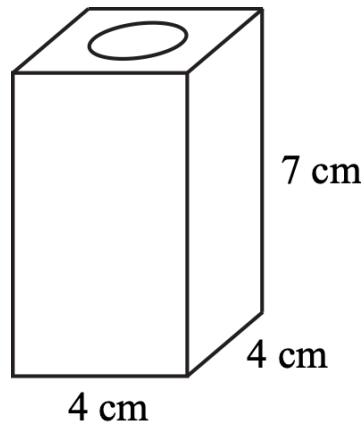
Solution:

Given that n has a remainder of 8 when divided by 10, then n is 8 more than a multiple of 10.

This means that n has a units digit of 8, which is 3 more than a units digit of 5, and thus n has a remainder of 3 when divided by 5.

We may confirm that examples of numbers that are equal to 8 more than a multiple of 10, such as 8, 18, 28, 38, do indeed leave a remainder of 3 when divided by 5.

16. A block of wood in the shape of a rectangular prism has length 4 cm, width 4 cm, and height 7 cm. A cylindrical hole with radius 1 cm is drilled through the centre, as shown.



To the nearest cm^3 , what is the volume of the block of wood after the hole is drilled? (Note: The volume of a cylinder with radius r and height h is $\pi r^2 h$.)

- (A) 90 cm^3 (B) 122 cm^3 (C) 106 cm^3 (D) 84 cm^3 (E) 92 cm^3

Source: 2024 Gauss Grade 8 #16

Primary Topics: Geometry and Measurement

Secondary Topics: Prisms | Cylinders | Volume

Answer: A

Solution:

Before the hole was drilled, the volume of the block of wood was $4 \text{ cm} \times 4 \text{ cm} \times 7 \text{ cm} = 112 \text{ cm}^3$.

The cylindrical hole has radius 1 cm and height equal to that of the block of wood, 7 cm.

The volume of the cylindrical hole is thus $\pi \times (1 \text{ cm})^2 \times 7 \text{ cm} = 7\pi \text{ cm}^3$.

In cm^3 , the volume of the block of wood after the hole is drilled is $112 - 7\pi \approx 90.01$, which when rounded to the nearest cm^3 , is 90 cm^3 .

17. The Gaussbot factory assembles robots. Each robot comes in one of three colours: red, blue, or green. Each robot also has a number stamped on its head: 1, 2, 3, or 4. The n th robot assembled is the first robot to have the same colour and the same number as a previously assembled robot. What is the greatest possible value of n ?
- (A) 11 (B) 12 (C) 13 (D) 7 (E) 8

Source: 2024 Gauss Grade 8 #17

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Patterning/Sequences/Series | Expressions | Factoring

Answer: C

Solution:

There are 3 different colours and 4 different numbers, and therefore $3 \times 4 = 12$ different kinds of robots that may be assembled. (These are R1, R2, R3, R4, B1, B2, B3, B4, G1, G2, G3, G4, where R, B, G represent the 3 colours red, blue, green.)

Since there are 12 different kinds of robots, it is possible that the first 12 robots assembled are all different from one another.

In this case, the 13th robot assembled would be the first robot to have the same colour and the same number as a previously assembled robot, and thus the greatest possible value of n is 13.

Notes:

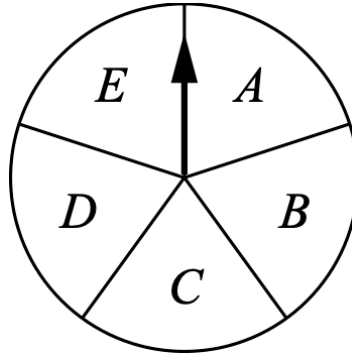
- i. It is possible that the first duplicate occurs earlier, but we want the latest that it can occur.
- ii. There must be a duplicate robot among the first 13 assembled, and so $n < 14$.
- iii. This solution makes use of the *Pigeonhole principle* – a concept worth further investigation.

It is possible that the first duplicate occurs earlier, but we want the latest that it can occur.

There must be a duplicate robot among the first 13 assembled, and so $n < 14$.

This solution makes use of the *Pigeonhole principle* – a concept worth further investigation.

18. A circular spinner is divided into five equal sections. An arrow is attached to the centre of the spinner and is positioned as shown.



The arrow is spun clockwise, and it stops in the section labelled *D*. Which of the following could have been the angle of rotation?

- (A) 530° (B) 550° (C) 630° (D) 675° (E) 700°

Source: 2024 Gauss Grade 8 #18

Primary Topics: Geometry and Measurement

Secondary Topics: Trigonometry | Angles

Answer: C

Solution:

The angle of one complete rotation measures 360° .

Since the spinner is divided into 5 equal sections, each $\frac{360^\circ}{5} = 72^\circ$ rotation moves the arrow to the next dividing line between two sections.

Rotating clockwise, the section labelled *D* is the 4th section.

This means that if the angle of rotation is greater than $3 \times 72^\circ = 216^\circ$ and less than $4 \times 72^\circ = 288^\circ$, then the arrow stops in the section labelled *D*.

Since each of the given answers is greater than 360° and less than 720° , the arrow must have spun through more than one and less than two complete rotations.

On the second spin around, the arrow will stop in the section labelled *D* if the angle of rotation is greater than $216^\circ + 360^\circ = 576^\circ$ and less than $288^\circ + 360^\circ = 648^\circ$.

Only one of the given answers lies in this range, and so the angle of rotation must be 630° .

19. Three *different* integers in a list have a mean (average) of 50 and a range of 14. What is the smallest possible integer in the list?
- (A) 40 (B) 43 (C) 39 (D) 42 (E) 41

Source: 2024 Gauss Grade 8 #19

Primary Topics: Data Analysis

Secondary Topics: Averages | Patterning/Sequences/Series

Answer: E

Solution:

Solution 1:

In this solution, we work backward from each of the given choices. Since we are asked to find the smallest possible integer in the list, we begin with the smallest of the five choices, 39.

If the smallest integer in the list is 39, then the largest integer in the list is $39 + 14 = 53$ (since the three integers have a range of 14).

If the three integers have a mean of 50, then they have a sum of $50 \times 3 = 150$. Two of the integers are 39 and 53, and so the third (the middle) integer is $150 - 39 - 53 = 58$.

Since 58 is greater than 53, this is not possible (the range of these three integers is $58 - 39 = 19$, not 14).

If the smallest integer in the list is 40 (the next smallest answer given), then the largest integer in the list is $40 + 14 = 54$.

If two of the integers are 40 and 54, then the third (the middle) integer is $150 - 40 - 54 = 56$.

Since 56 is greater than 54, this is not possible (the range of these three integers is $56 - 40 = 16$, not 14).

If the smallest integer in the list is 41 (the next smallest answer given), then the largest integer in the list is $41 + 14 = 55$.

If two of the integers are 41 and 55, then the third (the middle) integer is $150 - 41 - 55 = 54$.

We may confirm that the three integers 41, 54, 55 indeed have a range of 14 and a mean of 50.

We have shown that 41 is the smallest of the five choices to satisfy the given conditions, and so 41 is the smallest possible integer in the list.

Solution 2:

Assume that the list of 3 integers, ordered from smallest to largest, is a, b, c .

Since a is the smallest integer in the list and c is the largest, and the list has a range of 14, then c is 14 more than a or $c = a + 14$.

The three integers have a mean of 50, and so $\frac{a+b+c}{3} = 50$ or $a + b + c = 150$.

Substituting $c = a + 14$, the previous equation becomes $a + b + a + 14 = 150$ or $2a + b = 136$.

To find the smallest possible value of a , we determine the largest possible value of b , recalling that $a < b$ and $b < c$ and so $b < a + 14$.

Since $2a$ is even for all possible values of a , and 136 is even, then b must be even (since $2a + b = 136$).

We begin by choosing an arbitrary value of b and using this value to determine a and c .

If $b = 52$, then $2a = 136 - 52 = 84$ and so $a = 42$ and $c = 42 + 14 = 56$.

In this case, the three integers are 42, 52, 56.

We continue to increase the value of b in order to determine the smallest possible value for a .

If $b = 54$, then $2a = 136 - 54 = 82$ and so $a = 41$ and $c = 41 + 14 = 55$.

In this case, the three integers are 41, 54, 55.

If $b = 56$, then $2a = 136 - 56 = 80$ and so $a = 40$ and $c = 40 + 14 = 54$.

In this case, the three integers are 40, 56, 54, which is not possible since $56 > 54$.

Continuing to increase the value of b will continue to give values of c that are less than b .

Decreasing the value of b will give values of a that are greater than 41.

Therefore, the smallest possible integer in the list is 41.

20. Kiran has a box containing three different types of fruit: apples, pears, and bananas. In the box, 21 pieces of fruit are not apples, 25 pieces of fruit are not pears, and 28 pieces of fruit are not bananas. How many pieces of fruit are in the box?
- (A) 53 (B) 32 (C) 46 (D) 37 (E) 51

Source: 2024 Gauss Grade 8 #20

Primary Topics: Counting and Probability

Secondary Topics: Counting | Logic

Answer: D

Solution:

Since 21 pieces of fruit are not apples, then the number of pears added to the number of bananas is 21

.

Since 25 pieces of fruit are not pears, then the number of apples added to the number of bananas is 25

.

Since 28 pieces of fruit are not bananas, then the number of apples added to the number of pears is 28

.

In adding these three totals $21 + 25 + 28$, we are counting the number of pears twice, the number of bananas twice, and the number of apples twice (since each appears in exactly two of the three statements above).

That is, $21 + 25 + 28 = 74$ is twice the total number of pieces of fruit, and so the number of pieces of fruit in the box is $\frac{74}{2} = 37$.

21. The prime factorization of $6 \times 5 \times 4 \times 3 \times 2 \times 1$ can be written in the form $2^a \times 3^b \times 5^c$. The value of $a + b + c$ is
- (A) 6 (B) 7 (C) 9 (D) 5 (E) 8

Source: 2024 Gauss Grade 8 #21

Primary Topics: Number Sense

Secondary Topics: Exponents | Operations | Factoring

Answer: B

Solution:

Expressing the product $6 \times 5 \times 4 \times 3 \times 2 \times 1$ in terms of prime factors, we get

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 = (2 \times 3) \times 5 \times (2 \times 2) \times 3 \times 2$$

This product has 4 factors of 2, 2 factors of 3, and 1 factor of 5, and thus is equal to $2^4 \times 3^2 \times 5^1$.

The value of $a + b + c$ is $4 + 2 + 1 = 7$.

22. The ratio of the number of quarters (worth \$0.25 each) to the number of dimes (worth \$0.10 each) to the number of nickels (worth \$0.05 each) in a jar is $9 : 3 : 2$. If the total value of the quarters and dimes is \$17.85, what is the total value of the nickels?
- (A) \$0.45 (B) \$0.50 (C) \$0.70 (D) \$0.35 (E) \$0.55

Source: 2024 Gauss Grade 8 #22

Primary Topics: Other | Number Sense

Secondary Topics: Equations Solving | Counting

Answer: C

Solution:

The ratio of the number of quarters to the number of dimes to the number of nickels is $9 : 3 : 2$.

This means that for some positive integer k , the number of quarters is $9k$, the number of dimes is $3k$, and the number of nickels is $2k$.

The value of each quarter is \$0.25 or 25 cents, and so the value of the quarters in the jar, in cents, is $25 \times 9k = 225k$.

The value of each dime is \$0.10 or 10 cents, and so the value of the dimes in the jar, in cents, is $10 \times 3k = 30k$.

The total value of the quarters and dimes is \$17.85 or 1785 cents, and so $225k + 30k = 1785$ or $255k = 1785$, and so $k = \frac{1785}{255} = 7$.

The number of nickels is $2k = 14$, and so the value of the nickels is $0.05 \times 14 = \$0.70$.

23. Five *different* integers, each greater than 0, have a sum of 264. The greatest common divisor of these five positive integers is d . What is the sum of the digits of the largest possible value of d ?

- (A) 4 (B) 3 (C) 8 (D) 2 (E) 6

Source: 2024 Gauss Grade 8 #23

Primary Topics: Number Sense

Secondary Topics: Counting | Operations | Factoring

Answer: B

Solution:

The smallest five positive integers, each having a divisor of d , are d , $2d$, $3d$, $4d$, and $5d$.

Thus, the smallest possible sum of five different positive integers whose greatest common divisor is d is $d + 2d + 3d + 4d + 5d = 15d$.

We know that the sum is at least $15d$ and is equal to 264, which means that $15d \leq 264$, and so $d \leq \frac{264}{15}$ or $d \leq 17.6$.

Each of the five integers is divisible by d , and so the sum of the five integers, 264, is divisible by d .

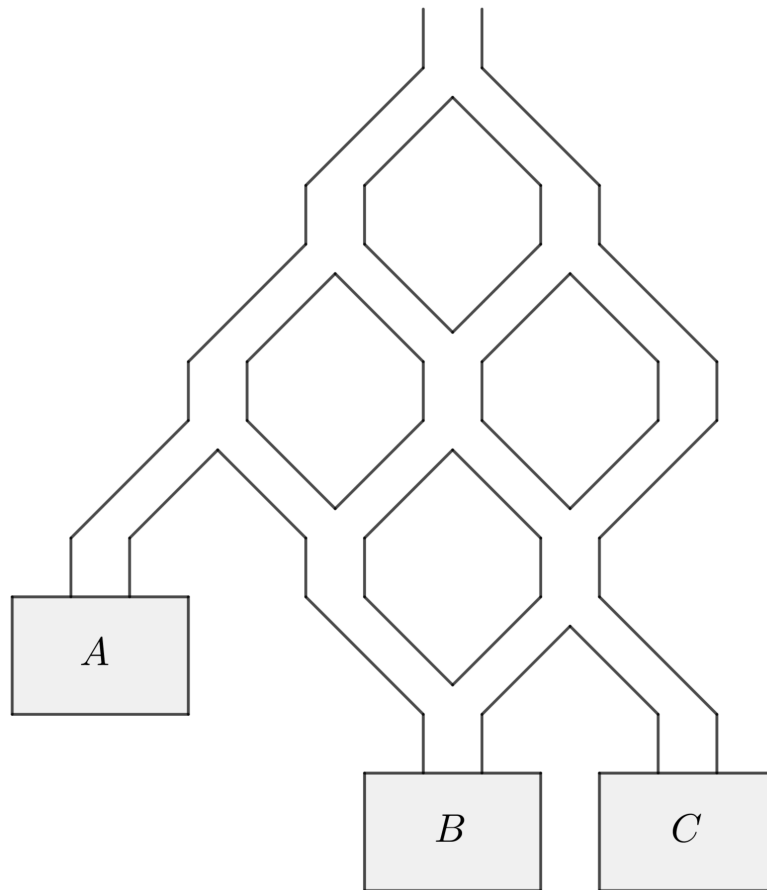
Thus, we want the largest possible divisor of 264 that is less than or equal to 17.

Since $264 = 2^3 \times 3 \times 11$, the divisors of 264 that are less than or equal to 17 are: 1, 2, 3, 4, 6, 8, 11, and 12, and so the largest possible value of d is 12.

The sum of the digits of the largest possible value of d is $1 + 2 = 3$.

(We note that 12, 24, 36, 48, and 144 are five such integers whose greatest common divisor is 12 and whose sum is 264.)

24. A network of pathways lead from a single opening to three bins, labelled A , B , C as shown. If a ball is dropped into the opening, it will follow a path and land in one of the bins. Every time a path splits, it is equally likely for the ball to follow either of the downward paths.



Ellen drops two balls, one after the other, into the opening. What is the probability that the two balls land in different bins?

- (A) $\frac{17}{32}$ (B) $\frac{27}{50}$ (C) $\frac{25}{64}$ (D) $\frac{1}{3}$ (E) $\frac{15}{32}$

Source: 2024 Gauss Grade 8 #24

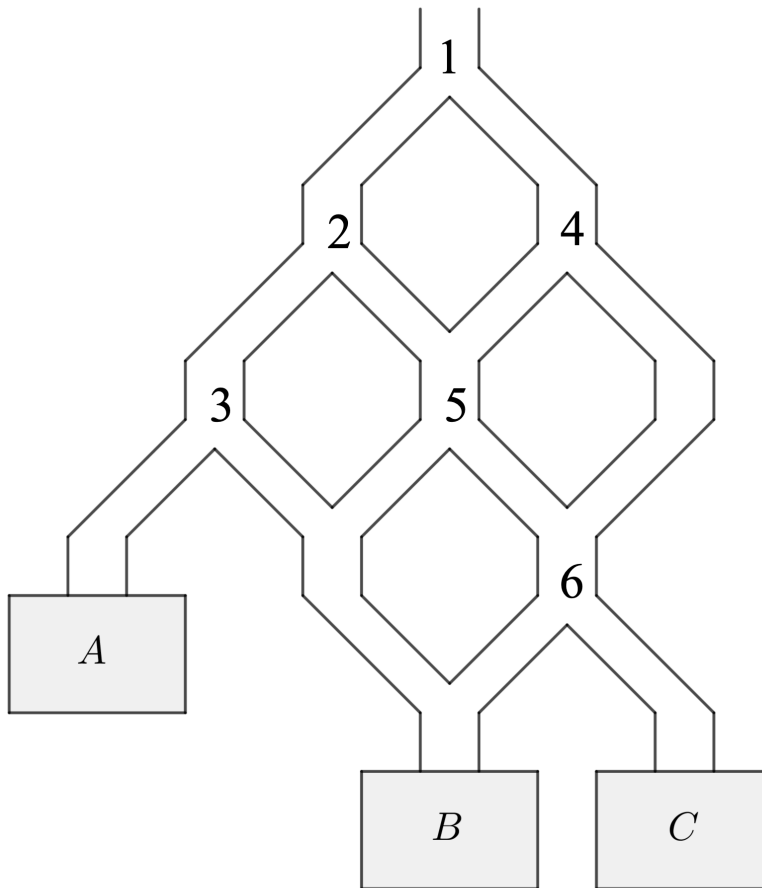
Primary Topics: Counting and Probability

Secondary Topics: Probability | Fractions/Ratios

Answer: A

Solution:

There are 6 different locations at which the path splits, and we label these splits 1 to 6, as shown.



We begin by determining the probability that a ball lands in the bin labelled *A*.

There is exactly one path that leads to bin *A*.

This path travels downward to the left at each of the three splits labelled 1, 2 and 3.

At each of these splits, the probability that a ball travels to the left is $\frac{1}{2}$, and so the probability that a ball lands in bin *A* is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.

Next, we determine the probability that a ball lands in the bin labelled *C*.

There are exactly three paths that lead to bin *C*.

One of these paths travels downward to the right at each of the three splits labelled 1, 4 and 6.

Thus, the probability that a ball lands in bin *C* by following this path is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.

A second path to bin *C* travels downward to the right at split 1, to the left at split 4, to the right at split 5, and to the right at split 6.

The probability that a ball follows this path is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$.

The third and final path to bin *C* travels left at split 1, and to the right at each of the three splits 2, 5 and 6.

The probability that a ball follows this path is also $\frac{1}{16}$.

The probability that a ball lands in bin *C* is the sum of the probabilities of travelling each of these three paths or

$$\frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{2+1+1}{16} = \frac{4}{16} = \frac{1}{4}$$

Finally, we determine the probability that a ball lands in bin *B*.

There are six different paths that lead to bin *B*, and we could determine the probability that a ball follows each of these just as we did for bins *A* and *C*.

However, it is more efficient to recognize that a ball must land in one of the three bins, and thus the probability that it lands in bin B is 1 minus the probability that it lands in bin A minus the probability that it lands in bin C , or

$$1 - \frac{1}{8} - \frac{1}{4} = \frac{8-1-2}{8} = \frac{5}{8}$$

The probability that the two balls land in different bins is equal to 1 minus the probability that the two balls land in the same bin.

The probability that a ball lands in bin A is $\frac{1}{8}$, and so the probability that two balls land in bin A is $\frac{1}{8} \times \frac{1}{8} = \frac{1}{64}$.

The probability that a ball lands in bin C is $\frac{1}{4}$, and so the probability that two balls land in bin C is $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$.

The probability that a ball lands in bin B is $\frac{5}{8}$, and so the probability that two balls land in bin B is $\frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$.

Therefore, the probability that the two balls land in different bins is equal to

$$1 - \frac{1}{64} - \frac{1}{16} - \frac{25}{64} = \frac{64-1-4-25}{64} = \frac{34}{64} = \frac{17}{32}$$

From split 1, moving to the left leads to split 2 and moving to the right leads to split 4.

From 2, left leads to 3 and right leads to 5.

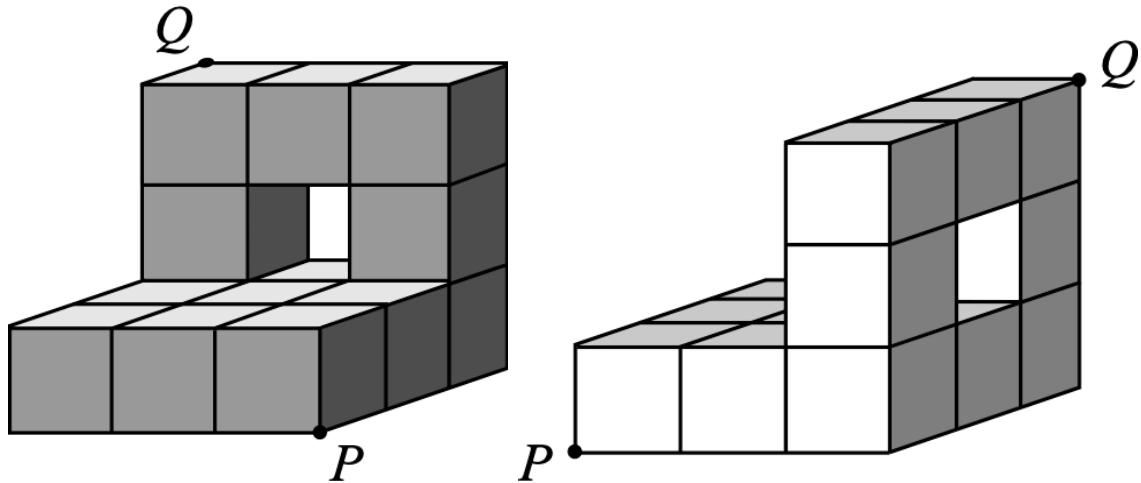
From 4, left leads to 5 and right leads to 6.

From 3, left leads to bin A and right leads to bin B.

From 5, left leads to bin B and right leads to split 6.

From 6, left leads to bin B and right leads to bin C.

25. A figure is constructed using fourteen $1 \times 1 \times 1$ cubes. Nine of the $1 \times 1 \times 1$ cubes are used to make the bottom layer and five additional $1 \times 1 \times 1$ cubes are positioned on top of the bottom layer. The figure is shown from two different perspectives.



First perspective

9 cubes form a 3 by 3 base layer of the figure, arranged into a front row, a middle row, and a back row.

5 additional cubes are placed above the back row forming a shape like a rectangular arch: 2 cubes are stacked on top of both the leftmost cube and the rightmost cube, forming a middle layer and a top layer, and 1 cube is placed between the 2 top cubes completing a row of 3 cubes in the top layer.

Points P and Q mark opposite corners of the figure: P is located at the bottom-right corner of the front face and Q is located at the top-left corner of the back face.

Second perspective

First perspective rotated ninety degrees so that the right face in the first view becomes the front face in the second view. In this view, P is located at the bottom-left corner of the front face and Q is located at the top-right corner of the back face.

An ant begins at P and walks a distance d on the surface of the figure to arrive at Q . The smallest possible value of d is closest to

- (A) 6.43 (B) 6.40 (C) 6.71 (D) 6.66 (E) 6.48

Source: 2024 Gauss Grade 8 #25

Primary Topics: Geometry and Measurement

Secondary Topics: Polygons

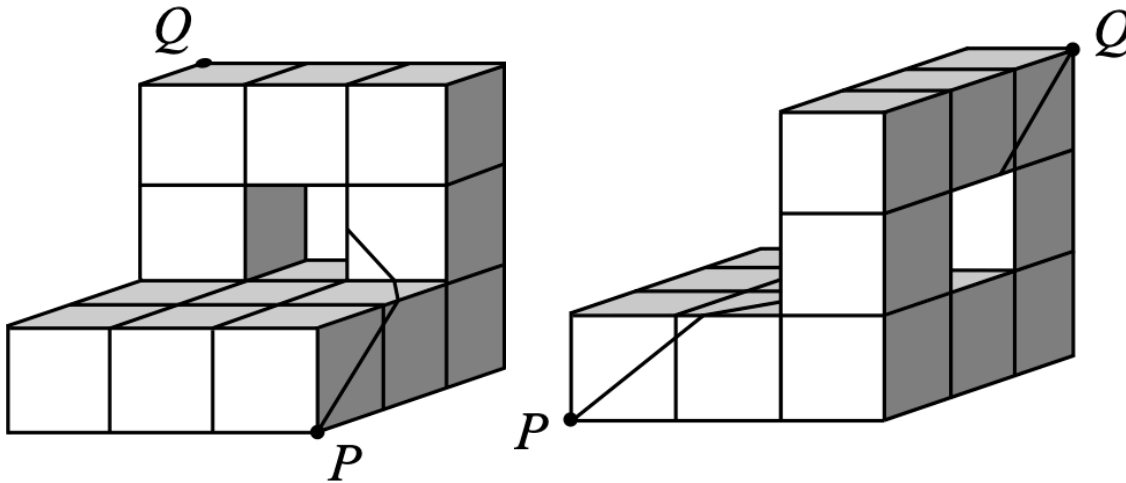
Answer: B

Solution:

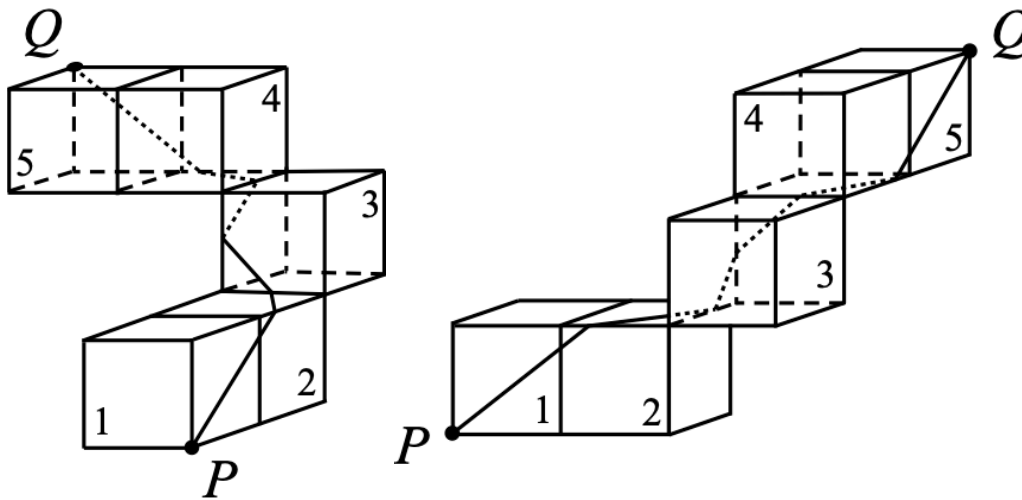
The smallest possible value of d is closest to 6.40.

On a flat surface, the shortest distance between two points is along a straight line between the points. The ant must walk on the surface of the figure, and so to determine a straight line distance between P and Q , we can "flatten" the figure.

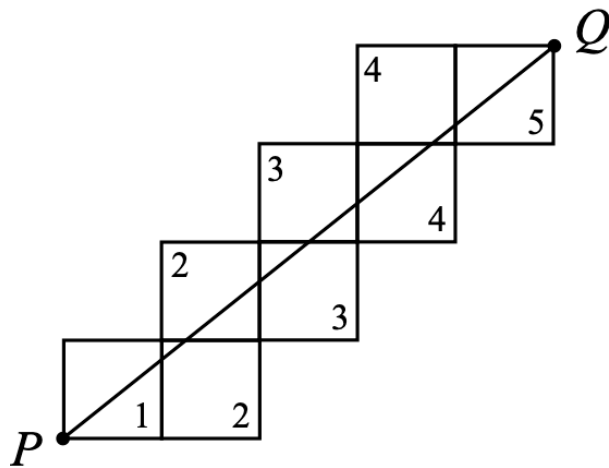
In the figures below, we show a straight line path whose distance is closest to 6.40 from each of the two perspectives.



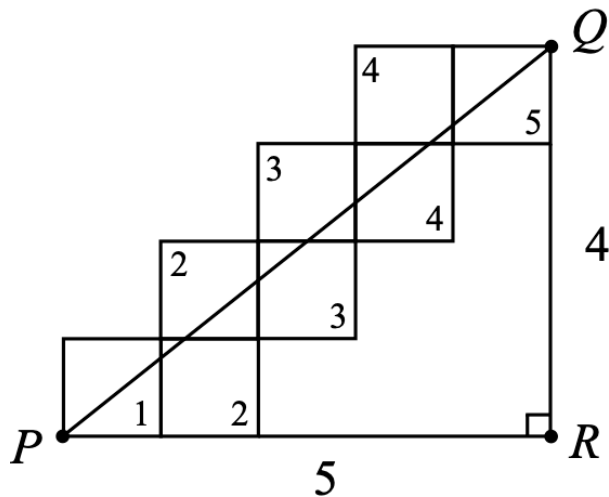
To help see this path more clearly, we strip away all but the 5 cubes that the ant walks on, below.



To see that this path is along a straight line from P to Q , we draw a partial net of the previous diagrams. This partial net includes each face that the ant walks on. The numbers shown in the diagram below indicate that the face is from the cube with the matching number in the diagrams above.



To determine the length of PQ , we position R so that PR is perpendicular to QR , as shown below.



Triangle PQR is a right-angled triangle with $PR = 5$ and $QR = 4$, and so by the Pythagorean Theorem, we get $PQ = \sqrt{5^2 + 4^2} = \sqrt{41}$, and so the distance d is closest to 6.40. Since 6.40 is the smallest of the five choices given, and we have shown that there is a path of this length from P to Q on the figure's surface, then the smallest possible value of d is 6.40.

Each of the other four given answers, 6.43, 6.48, 6.66, and 6.71 is a result of the ant travelling along other paths from P to Q . For example, there exists a second possible straight line path on a net of this figure for which PQ is approximately 6.71. Can you determine this path, and each of the other three paths from P to Q whose lengths are equal to 6.43, 6.48, and 6.66? There are also paths different from the one shown above for which $PQ = \sqrt{41}$. Can you find these?

In the bottom layer, the right face of the rightmost cube in the front row, the right face of the rightmost cube in the middle row, then the top face of the rightmost cube in the middle row, and in the middle layer, the front face of the rightmost cube.